

Complete Mathematical Foundations & Derivations Guide

A Comprehensive, High-School-to-Advanced Reference for
Zero Forcing Numbers of Cubic Graphs with Cyclic Symmetry

Research Reference Manual

Graph Theory • Linear Algebra • Combinatorial Optimisation
Fourier Analysis on Finite Groups

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Abstract

This pedagogical reference explains **every mathematical concept, equation, variable, and proof** used in the research project “A Rotation-Bootstrap Method for Zero Forcing Bounds on Cubic Graphs with Cyclic Symmetry”. Starting from high-school foundations (graphs, matrices, eigenvalues, modular arithmetic, and complex roots of unity), each object is built from first principles with intuitive analogies, precise definitions, step-by-step derivations, and fully worked numerical examples. It covers the zero forcing process, the maximum nullity connection, forts and hitting-set duality, the rotation bootstrap, Fourier block-diagonalisation of circulant structures, and Chebyshev certificates for lower bounds.

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1 High School Foundation Primer

1.1 Graphs

Definition 1.1: Graph

A **graph** $G = (V, E)$ consists of a set of **vertices** V (dots) and a set of **edges** E (lines joining pairs of dots). We write $|V| = N$ for the number of vertices.

- **Neighbour:** u and v are neighbours if $uv \in E$. The **neighbourhood** $N(v)$ is the set of all neighbours of v .
- **Degree** $\deg(v) = |N(v)|$: how many edges touch v .
- **Cubic graph:** every vertex has degree exactly 3.

Intuition: what is a cubic graph?

Picture a road network where *every* intersection has exactly three roads meeting at it — no more, no less. Every graph in this project is cubic. That uniformity is what makes the algebra tractable: each vertex has three neighbours, so “all but one neighbour” always means “two of them”.

1.2 Modular arithmetic

Definition 1.2: Arithmetic mod n

Writing $a \equiv b \pmod{n}$ means a and b leave the same remainder when divided by n . We label vertices $0, 1, \dots, n-1$ and treat index n as 0, index $n+1$ as 1, and so on — arithmetic “wraps around” a circle.

The **greatest common divisor** $\gcd(n, k)$ is the largest integer dividing both n and k . If $\gcd(n, k) = 1$ we call n and k **coprime**.

Intuition: why gcd decides the number of cycles

Stand on a clock with n hours and repeatedly jump k hours forward. Do you visit every hour before returning to the start?

- $n = 12, k = 5$: $0 \rightarrow 5 \rightarrow 10 \rightarrow 3 \rightarrow 8 \rightarrow 1 \rightarrow \dots$ You hit all 12 — one big loop. Here $\gcd(12, 5) = 1$.
- $n = 12, k = 4$: $0 \rightarrow 4 \rightarrow 8 \rightarrow 0$. You are stuck in a 3-loop and never reach 1. Here $\gcd(12, 4) = 4$, and the 12 hours split into 4 separate loops of length 3.

Rule: jumping by k on an n -cycle produces exactly $d = \gcd(n, k)$ separate loops, each of length n/d . This single fact drives much of the project.

1.3 Linear algebra essentials

Definition 1.3: Matrices, nullity, eigenvalues

- A **matrix** A is a square grid of numbers; A_{uv} is the entry in row u , column v . It is **symmetric** if $A_{uv} = A_{vu}$.
- The **null space** (kernel) is $\ker(A) = \{x : Ax = \mathbf{0}\}$ — all vectors the matrix crushes to zero. Its dimension is the **nullity**, $\text{null}(A)$.
- The **rank** is the dimension of the output space, and **rank-nullity** says $\text{rank}(A) + \text{null}(A) = N$.
- λ is an **eigenvalue** with **eigenvector** $x \neq 0$ if $Ax = \lambda x$: the matrix only stretches x , never turns it.
- For a symmetric matrix all eigenvalues are real, and

$$\text{null}(A) = \#\{\text{eigenvalues equal to } 0\}.$$

Intuition: nullity counts “directions that vanish”

A matrix is a machine turning vectors into vectors. Nullity counts how many independent input directions come out as zero — how much information the machine destroys. **High nullity means a very “lossy” machine.** We will want to build the lossiest possible machine whose wiring diagram is our graph.

1.4 Complex numbers and roots of unity

Definition 1.4: Roots of unity

The **imaginary unit** satisfies $i^2 = -1$. Euler’s formula says $e^{i\theta} = \cos \theta + i \sin \theta$, a point on the unit circle at angle θ . The **n -th roots of unity** are

$$\zeta_\ell = e^{2\pi i \ell / n}, \quad \ell = 0, 1, \dots, n-1,$$

the n points evenly spaced around the unit circle. They satisfy $\zeta_\ell^n = 1$, and crucially

$$\zeta_\ell + \zeta_\ell^{-1} = 2 \cos\left(\frac{2\pi \ell}{n}\right),$$

which is a *real* number even though each term is complex.

Intuition: why roots of unity appear at all

Our graphs look identical after rotating by one step. Roots of unity are the algebraic encoding of “rotating by one step”: multiplying by ζ shifts you one notch around the circle. Because our matrices commute with rotation, they become *simple* in coordinates built from roots of unity — a huge matrix collapses into many tiny ones. This is exactly the idea behind Fourier analysis.

2 Zero Forcing: The Central Process

2.1 The colour change rule

Definition 2.1: Zero forcing

Colour some vertices **black** (filled) and the rest **white**.

Colour change rule. If a black vertex has *exactly one* white neighbour, that neighbour turns black. (We say the black vertex *forces* it.)

Apply the rule repeatedly until nothing more changes. The resulting black set is the **closure** $\text{cl}(S)$ of the starting set S . If $\text{cl}(S) = V$, then S is a **zero forcing set**. The **zero forcing number** $Z(G)$ is the size of the smallest one.

Intuition: the rumour with one listener

A black vertex knows a rumour. It will only whisper if it has exactly one person left who hasn't heard — with two or more uninformed neighbours it cannot decide whom to tell, so it stays silent. $Z(G)$ asks: what is the *fewest* people who must start knowing the rumour for it to reach everybody?

Note the strangeness: having *more* black neighbours can make a vertex *less* useful, because it needs exactly one white neighbour to act.

Three properties of closure we use constantly

1. **Monotone:** if $S \subseteq T$ then $\text{cl}(S) \subseteq \text{cl}(T)$. Starting with more never finishes with less.
2. **Idempotent:** $\text{cl}(\text{cl}(S)) = \text{cl}(S)$. Once the process halts, restarting it changes nothing.
3. **Automorphism-equivariant:** if ρ is a symmetry of the graph then $\rho(\text{cl}(S)) = \text{cl}(\rho(S))$. Rotating first and then forcing gives the same answer as forcing first and then rotating.

2.2 Why anyone cares

Definition 2.2: Pattern matrices, minimum rank, maximum nullity

Let $S(G)$ be all real symmetric matrices A whose off-diagonal pattern *is* the graph:

$$A_{uv} \neq 0 \iff uv \in E(G) \quad (u \neq v),$$

with the diagonal entries completely free. The **maximum nullity** is

$$M(G) = \max\{\text{null}(A) : A \in S(G)\}.$$

Theorem 2.1 (the reason zero forcing exists)

For every graph G ,

$$M(G) \leq Z(G).$$

Intuition: why forcing bounds nullity

Suppose $Ax = 0$ and we know x is zero on a zero forcing set S . Take a black vertex u with exactly one white neighbour w . Row u of $Ax = 0$ reads

$$A_{uu}x_u + \sum_{v \in N(u)} A_{uv}x_v = 0.$$

Every term is zero except possibly $A_{uw}x_w$ — because u and all its other neighbours are black, hence zero. So $A_{uw}x_w = 0$, and since $A_{uw} \neq 0$ we get $x_w = 0$. The zero *propagates exactly like the colour change rule*. If S forces the whole graph, $x = 0$ everywhere, so a null vector is determined by its values on S , giving $\text{null}(A) \leq |S|$.

This single inequality is the engine of our lower bounds: to prove $Z(G) \geq r$, it suffices to *exhibit one matrix* in $S(G)$ with nullity r .

Where this shows up in the real world

- **Quantum control:** which subsystems must you be able to act on to steer the entire system? The answer is a zero forcing set.
- **Power grids:** where do you place phasor measurement units so the whole network's state is observable? Same combinatorial rule.
- **Minimum rank problem:** $M(G)$ is a long-studied invariant, and $Z(G)$ is its most useful computable bound.

3 The Graph Families

Definition 3.1: Generalized Petersen graph $P(n, k)$

Vertices: an **outer** ring $u_0, \dots, u_{n-1}$ and an **inner** ring $v_0, \dots, v_{n-1}$ ($2n$ total). Edges, all indices mod n :

$$\underbrace{u_i u_{i+1}}_{\text{outer cycle}}, \quad \underbrace{v_i v_{i+k}}_{\text{inner jumps}}, \quad \underbrace{u_i v_i}_{\text{spokes}}.$$

Each vertex has degree 3, so $P(n, k)$ is cubic. $P(5, 2)$ is the famous Petersen graph.

Intuition: a wheel with crossed spokes

Think of a bicycle wheel. The outer ring is the tyre, connected in a simple circle. The spokes run inward. But the inner ring is *not* a simple circle: vertex v_i connects to v_{i+k} , jumping k places. By the clock fact in §1.2, the inner vertices form $\gcd(n, k)$ separate loops. For $k = 1$ you get one inner circle (a prism); for larger k you get a star polygon.

Definition 3.2: I-graphs and double generalized Petersen graphs

- $I(n, j, k)$: same as above but the *outer* ring also jumps: $u_i u_{i+j}$, $v_i v_{i+k}$, $u_i v_i$. So $P(n, k) = I(n, 1, k)$, and the outer vertices form $\gcd(n, j)$ loops.
- $DP(n, k)$: *two* outer cycles $u_i u_{i+1}$ and $x_i x_{i+1}$, inner edges $w_i y_{i+k}$ and $y_i w_{i+k}$, spokes $u_i w_i$

and $x_i y_i$. It has $4n$ vertices and is also cubic.

Definition 3.3: The rotation automorphism

An **automorphism** is a relabelling of vertices that preserves all edges. Every family above admits the **rotation**

$$\rho : i \mapsto i + 1 \pmod{n}$$

applied to every index. Turning the whole picture one notch gives back the same graph.

4 Upper Bounds: The Rotation Bootstrap

The central trick of the project. It converts *one finite computation* into a theorem valid for *all* n .

Lemma 4.1 (Bootstrap)

Let ρ be an automorphism of a finite graph G , and suppose $\rho(S) \subseteq \text{cl}(S)$. Then $\rho(\text{cl}(S)) = \text{cl}(S)$.

Proof, one step at a time

$$\begin{aligned} \rho(\text{cl}(S)) &= \text{cl}(\rho(S)) && \text{closure commutes with symmetries} && (1) \\ &\subseteq \text{cl}(\text{cl}(S)) && \text{because } \rho(S) \subseteq \text{cl}(S), \text{ and closure is monotone} && (2) \\ &= \text{cl}(S) && \text{closure is idempotent} && (3) \end{aligned}$$

So $\rho(\text{cl}(S)) \subseteq \text{cl}(S)$. But ρ merely *relabels* a finite set, so both sides have the same number of elements. A subset of the same size must be the whole thing, hence equality. $\square$

Intuition: why “one notch” gives you everything

Suppose the closure of S contains a rotated copy of S . Then the closure cannot tell the difference between itself and its own rotation — it is *rotation-invariant*. And a rotation-invariant set containing u_0 must contain u_1 (rotate once), then u_2 (rotate again), and so on around the entire ring.

This is the whole trick: check a handful of forcing steps by hand, and rotation hands you the infinitely many remaining ones for free.

Lemma 4.2 (Closing)

Suppose the vertex set splits as outer O and inner I , each inner vertex has exactly one outer neighbour (its spoke) and vice versa, and every other neighbour of an outer vertex is outer. If $O \subseteq \text{cl}(S)$ then $\text{cl}(S) = V$.

Proof

Let $w \in I$ be unfilled, with outer neighbour o . The other two neighbours of o are outer, hence already filled. So o is black with exactly one white neighbour, namely w — and the colour change rule forces w . Therefore no unfilled inner vertex can survive. $\square$

Theorem 4.3 (I-graph bound)

For all $j, k \geq 1$ and $n \geq 2(j + k) + 1$,

$$Z(I(n, j, k)) \leq 2(j + k).$$

Taking $j = 1$ gives $Z(P(n, k)) \leq 2k + 2$.

Proof: a three-step forcing chain

Put $m = 2(j + k)$ and $S = \{u_0, u_1, \dots, u_{m-1}\}$, so $|S| = m$. Since $n \geq m + 1$, the indices $0, \dots, m$ are all different mod n .

Step 1 (outer forces its spoke). The outer neighbours of u_i are u_{i-j} and u_{i+j} . Both lie in S exactly when $j \leq i \leq m - 1 - j$. For such i , the only remaining neighbour is v_i , so u_i forces v_i . This fills $v_j, \dots, v_{m-1-j}$, i.e. $v_j, \dots, v_{j+2k-1}$.

Step 2 (inner reaches further). Look at v_{j+k} . Its three neighbours are $u_{j+k} \in S$; v_j , filled in Step 1; and v_{j+2k} , which is *not* filled (Step 1 stopped at index $j + 2k - 1$). One white neighbour, so v_{j+k} forces v_{j+2k} .

Step 3 (back out to the next outer vertex). Look at $u_{j+2k} = u_{m-j}$. Its neighbours are $u_{2k} \in S$; v_{j+2k} , filled in Step 2; and u_m , unfilled. So u_{j+2k} forces u_m .

Conclusion. We now have $u_1, \dots, u_{m-1}$ (in S) and u_m (just forced), i.e. $\rho(S) \subseteq \text{cl}(S)$. Lemma 4.1 makes $\text{cl}(S)$ rotation-invariant; it contains u_0 , whose rotations sweep out every outer vertex; Lemma 4.2 then fills the inner ring. So S is a zero forcing set of size m . $\square$

Why $m = 2(j + k)$ exactly?

Step 2 needs v_{m-j-2k} to be among the vertices filled in Step 1, whose first index is j . Setting $m = 2(j + k)$ makes $m - j - 2k = j$ *precisely*. The number is not guessed — it is the smallest choice that lets the chain close.

Theorem 4.4 (Double generalized Petersen bound)

For all $k \geq 1$ and $n \geq 2k + 3$, $Z(DP(n, k)) \leq 4k + 4$, witnessed by $S = \{u_0, \dots, u_{2k+1}\} \cup \{x_0, \dots, x_{2k+1}\}$. The proof is the same chain run in parallel on both outer cycles. Our computations show the bound is **attained** at $k = 1$ and $k = 2$.

5 Forts: Why Lower Bounds Are Hard

Definition 5.1: Fort

A **fort** is a non-empty set F of vertices such that every vertex *outside* F has either 0 or at least 2 neighbours inside F .

Intuition: a fort is an unbreakable stronghold

Imagine F is entirely white. Can anyone outside force their way in? A vertex $x \notin F$ can only force if it has exactly one white neighbour. But the fort condition says x has 0 or ≥ 2 neighbours in F — never exactly one. With 0 it cannot reach the fort at all; with ≥ 2 it is paralysed by indecision. **So a fort left entirely white stays white forever.**

Theorem 5.2 (Fort duality)

S is a zero forcing set *if and only if* S intersects every fort. Consequently $Z(G)$ equals the minimum number of vertices meeting all forts — a **hitting set** problem.

Three lower-bound attempts, and why each failed

1. **Disjoint forts.** If t forts share no vertices, a forcing set needs a separate vertex in each, so $Z \geq t$. *Failed:* forts in $P(n, k)$ have 7–13 vertices while the graph has only $2n \leq 56$, so at most *two* fit disjointly. Bound: $Z \geq 2$. Useless.
 2. **The LP relaxation.** Allow fractional vertices. *Failed:* measured $\text{LP}/Z \approx 0.43\text{--}0.50$ — an integrality gap near 2 that widens with k . This also explains why exact solvers crawl: branch-and-bound must close a factor-of-two gap by brute force.
 3. **Spectral (Hoffman-type) bound.** *Failed:* gives 1.2–2.9 against true Z of 6–12.
- All three are combinatorial, and all three degrade as k grows.** That is what motivated changing the *category* of argument entirely.

6 Lower Bounds by Matrix Certificates

6.1 The strategy

By Theorem 2.1, $Z(G) \geq M(G)$. So to prove $Z(G) \geq r$ we need only **build one matrix** in $S(G)$ with nullity r . The difficulty: A is $2n \times 2n$, which for $n = 28$ is 56×56 . Symmetry rescues us.

6.2 Fourier block-diagonalisation

Definition 6.1: Equivariant matrix

Let P be the cyclic shift (P sends coordinate i to $i + 1$). Build

$$U = aI + b(P + P^{-1}), \quad V = dI + e(P^k + P^{-k}), \quad C = cI,$$

and assemble $A = \begin{pmatrix} U & C \\ C & V \end{pmatrix}$, where U handles outer edges, V inner edges, C the spokes.

Requiring $b, c, e \neq 0$ makes the zero pattern exactly $P(n, k)$, so $A \in S(G)$. The diagonals

a, d are free.

Intuition: why symmetry shatters a big matrix into tiny ones

Because A treats every rotation identically, it cannot mix the “rotational frequencies”. Feed in a wave $x_i = \zeta^i$ where ζ is an n -th root of unity, and A returns a multiple of the same wave. Each frequency lives in its own private 2-dimensional world (outer value, inner value), so the $2n \times 2n$ matrix splits into n independent 2×2 blocks.

The high-school analogue: a symmetric object vibrates in independent modes, and analysing one mode at a time is far easier than the whole object at once.

The blocks

Because P has eigenvalue ζ^m on the m -th wave, $P + P^{-1}$ contributes $2 \cos(2\pi m/n)$. Writing

$$s_m = 2 \cos \frac{2\pi m}{n}, \quad t_m = 2 \cos \frac{2\pi km}{n},$$

the matrix becomes n blocks

$$M_m = \begin{pmatrix} a + b s_m & c \\ c & d + e t_m \end{pmatrix}, \quad \det M_m = (a + b s_m)(d + e t_m) - c^2.$$

A 2×2 matrix is singular exactly when its determinant is 0, so

$$\text{null}(A) = \#\{m : (a + b s_m)(d + e t_m) = c^2\}.$$

6.3 The linear trick

Making the condition linear

Set $b = 1$ and expand:

$$(a + s)(d + e t) - c^2 = e(st) + ds + (ae)t + (ad - c^2) = 0.$$

Read the four bracketed quantities (e , d , ae , $ad - c^2$) as *unknown coefficients*. Then the condition is a **linear, homogeneous** equation in them, and each point (s_m, t_m) gives one such equation. Three points determine the coefficients up to scale — a nullspace computation, *not* a search.

Why this matters so much

Solving exactly means the blocks are singular **by construction**, not by numerical luck. A numerical optimiser hunting for near-zero eigenvalues can be fooled; an exact linear solve cannot. (In this project an optimiser once reported nullity 7 where $Z = 6$ — impossible, since $Z \geq M \geq \text{nullity}$. It had shrunk an edge weight to 0.0003, numerically *deleting* an edge. The inequality caught the bug automatically.)

6.4 Chebyshev: from one n to all n

Definition 6.2: Chebyshev polynomials

T_k is the polynomial with $\cos(k\theta) = T_k(\cos\theta)$. For example $T_2(y) = 2y^2 - 1$ and $T_3(y) = 4y^3 - 3y$.

The key observation

Since $s = 2\cos\theta$ and $t = 2\cos k\theta$, we have *exactly*

$$t = 2T_k(s/2),$$

a polynomial in s of degree k . So the singularity condition is a **polynomial in s alone of degree $k + 1$** , and the singular blocks are precisely its roots.

Two consequences, both important

1. **Why the bound caps at 6.** For $k = 2$ the polynomial is a cubic, whose coefficients $(e, ae, D, aD - c^2)$ are freely choosable, so *any* three s -values can be made roots. Three roots, each counted twice (because m and $n - m$ give the same s), gives nullity 6 — and a cubic has no fourth root, so 6 is the ceiling.
2. **Uniformity.** The recipe (pick three s -values, form the monic cubic, read off $e = A_3$, $a = A_2/A_3$, $D = A_1$, $c^2 = aD - A_0$) works for *every* n , not one at a time.

Theorem 6.3 (what we proved)

- $Z(P(n, 2)) = 6$ for $11 \leq n \leq 41$ — both directions.
- Using period-2 matrices (commuting only with ρ^2 , giving ten parameters and 4×4 blocks): $Z(P(n, 4)) \geq 8$ for $n = 18$ and all even $22 \leq n \leq 40$, so $8 \leq Z(P(n, 4)) \leq 10$.
- $Z(P(24, 6)) \geq 10$.

No previously published lower bound on $Z(P(n, k))$ for $k \geq 3$ is known to us.

7 Complete Worked Numerical Example

We carry out *every* computation by hand on $P(6, 2)$: 12 vertices, outer $u_0..u_5$, inner $v_0..v_5$.

7.1 Step 1: the graph

Outer edges $u_i u_{i+1}$: $(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 0)$.

Inner edges $v_i v_{i+2}$: $(0, 2), (2, 4), (4, 0)$ and $(1, 3), (3, 5), (5, 1)$.

Spokes: $u_i v_i$ for each i .

Since $\gcd(6, 2) = 2$, the inner vertices form 2 triangles: $\{v_0, v_2, v_4\}$ and $\{v_1, v_3, v_5\}$ — exactly as the clock rule predicts.

Degrees: u_i has $u_{i\pm 1}$ and v_i ; v_i has $v_{i\pm 2}$ and u_i . All degree 3. ✓

7.2 Step 2: run the forcing rule by hand

Take $S = \{u_0, u_1, u_2, u_3, u_4, u_5\}$, the whole outer ring.

- u_0 's neighbours are u_5, u_1 (black) and v_0 (white) — exactly one white neighbour, so u_0 **forces** v_0 .
- Likewise each u_i forces v_i .

After one round every vertex is black, so $\text{cl}(S) = V$ and $Z(P(6, 2)) \leq 6$. This is Lemma 4.2 in miniature.

7.3 Step 3: check the bootstrap bound

Theorem 4.3 with $j = 1, k = 2$ gives $m = 2(1 + 2) = 6$ and requires $n \geq 7$. Our $n = 6$ is *below* the hypothesis — and indeed the chain degenerates, because $u_m = u_6 = u_0$ already lies in S . This is precisely why the theorem demands $n \geq 2(j + k) + 1$.

7.4 Step 4: the certificate blocks

With $n = 6, k = 2$:

m	0	1	2	3	4	5
$s_m = 2 \cos(2\pi m/6)$	2	1	-1	-2	-1	1
$t_m = 2 \cos(4\pi m/6)$	2	-1	-1	2	-1	-1

Check the Chebyshev identity $t = s^2 - 2$: for $m = 1, s = 1$ and $s^2 - 2 = -1 = t$. ✓ For $m = 3, s = -2$ and $s^2 - 2 = 2 = t$. ✓

Distinct points: $(2, 2), (1, -1), (-1, -1), (-2, 2)$ — only four. Choose the three roots $s \in \{1, -1, -2\}$ and form

$$(s - 1)(s + 1)(s + 2) = s^3 + 2s^2 - s - 2.$$

So $A_3 = 1, A_2 = 2, A_1 = -1, A_0 = -2$, giving

$$e = 1, \quad a = A_2/A_3 = 2, \quad D = A_1 = -1, \quad c^2 = aD - A_0 = (2)(-1) - (-2) = 0.$$

Here $c^2 = 0$, which is **not admissible** — a zero spoke weight would disconnect outer from inner and leave $S(G)$. This is the degeneracy discussed in the main paper, and it is why $n = 10$ also failed and why we fit *four* rather than five points in the period-2 construction. For larger n more distinct s -values are available and admissible triples do exist.

7.5 Step 5: verify $Z \geq M$ numerically

Building any admissible A and computing eigenvalues gives $\text{null}(A) \leq Z(P(6, 2))$ every time — the inequality that served as our automatic bug detector.

8 Master Table of Variables

Symbol	Name	Meaning
$G = (V, E)$	Graph	Vertices and edges
$N(v), \deg(v)$	Neighbourhood, degree	Neighbours of v ; how many
$\text{cl}(S)$	Closure	Result of applying the colour change rule to S until it halts
$Z(G)$	Zero forcing number	Size of the smallest zero forcing set
$S(G)$	Pattern class	Symmetric matrices whose off-diagonal nonzeros are exactly $E(G)$
$M(G)$	Maximum nullity	Largest nullity over $S(G)$; satisfies $M \leq Z$
$P(n, k)$	Generalized Petersen	Outer $u_i u_{i+1}$, inner $v_i v_{i+k}$, spokes $u_i v_i$
$I(n, j, k)$	I-graph	As above with outer step j ; $P(n, k) = I(n, 1, k)$
$DP(n, k)$	Double gen. Petersen	Two outer cycles, cross-linked inner edges, $4n$ vertices
ρ	Rotation	Automorphism $i \mapsto i + 1 \pmod n$
d_1, d_2	$\gcd(n, j), \gcd(n, k)$	Number of outer / inner cycles
F	Fort	Non-empty set every outside vertex meets in 0 or ≥ 2 places
P	Cyclic shift matrix	Sends coordinate i to $i + 1$
ζ	Root of unity	$e^{2\pi i \ell / n}$; encodes “rotate one notch”
s_m, t_m	Block parameters	$2 \cos(2\pi m/n)$ and $2 \cos(2\pi km/n)$
M_m	Fourier block	2×2 matrix whose determinant decides singularity
T_k	Chebyshev polynomial	$\cos k\theta = T_k(\cos \theta)$; gives $t = 2T_k(s/2)$
a, b, c, d, e	Matrix parameters	Outer diagonal, outer edge, spoke, inner diagonal, inner edge

9 Summary & Key Takeaways

The five ideas, in order

1. **Zero forcing** is a spreading rule with a quirky trigger: a filled vertex acts only when exactly one neighbour is unfilled.
2. $M(G) \leq Z(G)$ converts a combinatorial question into a matrix question. Upper bounds on Z come from exhibiting a forcing *set*; lower bounds come from exhibiting a *matrix*.
3. **The rotation bootstrap** turns a three-step hand computation into a theorem for all n : if the closure swallows one rotated copy of the seed, it is rotation-invariant, hence everything.
4. **Fourier** splits a $2n \times 2n$ matrix into n blocks of size two, and the singularity condition is *linear* in four coefficients — solvable exactly, so certificates are correct by construction.
5. **Chebyshev** ($t = 2T_k(s/2)$) makes the condition a polynomial of degree $k + 1$ in one variable. This explains the ceiling of 6 for ρ -equivariant matrices and makes the constructions uniform in n .

What is proved, and what is not

Proved: $Z(I(n, j, k)) \leq 2(j + k)$; $Z(DP(n, k)) \leq 4k + 4$ (sharp at $k = 1, 2$); $Z(P(n, 2)) = 6$ for $11 \leq n \leq 41$; $Z(P(n, 4)) \geq 8$ across a range, giving $8 \leq Z(P(n, 4)) \leq 10$.

Not proved: the matching lower bound $Z \geq 2k + 2$ in general. Our certificates reach within 2 of it but not equality. Three combinatorial techniques were shown insufficient *in principle*, not merely in practice.

Refuted along the way: a conjectured threshold $n_0(k) = 5k - 2$ (killed by $Z(P(29, 6)) = 12$) and a conjectured bound $Z \leq 2(\gcd(n, j) + \gcd(n, k))$ (killed by $Z(P(11, 3)) = 7 > 4$). Both were our own, and recording them is part of the method.